

## DSA 210 Final Report

### League of Legends Meta Analysis

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#### 1. Introduction

In competitive multiplayer online games, performance is influenced by multiple strategic and mechanical factors. One of the most widely discussed concepts in games such as *League of Legends* is the “meta,” which refers to champions, strategies, or playstyles that are perceived as stronger or more effective within a given competitive environment.

Players often prioritize meta champions because they are believed to offer a competitive advantage, particularly in ranked gameplay where win probability becomes increasingly important. However, while this assumption is commonly accepted within the gaming community, it is important to evaluate whether this advantage is supported by measurable data.

This project investigates whether meta champions provide a statistically meaningful advantage in match outcomes and whether this effect differs across player skill levels. By combining exploratory data analysis, statistical testing, and machine learning approaches, the study aims to understand the role of champion meta in determining win rates.

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#### 2. Research Question

This project was designed around the following primary research question:

**Do meta champions significantly increase win rates, and does this effect vary across different skill levels (ranks)?**

To answer this question, the project focused on three sub-objectives:

- Determining whether champions categorized as “meta” show higher average win rates.
- Evaluating whether this advantage differs between low-elo and high-elo players.
- Testing whether match outcomes can be predicted using champion-related and rank-based features.

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#### 3. Dataset

The analysis was conducted using multiple relational CSV tables related to *League of Legends* ranked gameplay.

The original dataset consisted of the following tables:

- **MatchTbl** → Match-level information
- **TeamMatchTbl** → Team-level match performance
- **SummonerMatchTbl** → Player participation records
- **ChampionTbl** → Champion metadata
- **RankTbl** → Rank information

These datasets were merged using relational keys such as:

- MatchId
- MatchFk
- ChampionFk
- RankFk

After filtering only ranked/classic matches and removing incomplete observations, the final processed dataset contained:

- **113,254 ranked/classic matches**
- **125,001 player-level observations**
- **113 reliable champions** (minimum 500-game threshold)
- **10 competitive rank categories (Iron to Challenger)**

This structure allowed both player-level and champion-level analysis.

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## 4. Data Preparation

A structured preprocessing pipeline was applied to prepare the dataset for analysis.

### 4.1 Data Merging

The relational tables were merged in the following order:

1. Match data with team-level match data
2. Player-level data with merged match data
3. Champion metadata integration
4. Rank metadata integration

This created a unified player-level analytical dataset.

## 4.2 Filtering

Only ranked and classic queue matches were included to ensure competitive consistency.

The following conditions were applied:

- QueueType = CLASSIC
- RankFk ≠ 0

This removed unranked or incomplete entries.

## 4.3 Feature Engineering

Several derived features were created:

- **Win** → Binary match outcome (1 = win, 0 = loss)
- **IsBlue** → Whether the player belonged to blue side
- **IsRed** → Whether the player belonged to red side
- **ChampionName** → Champion mapping
- **RankName** → Rank mapping

## 4.4 Meta Definition

Meta champions were defined based on average champion win rate.

A champion was classified as **meta** if:

**Champion Average Win Rate > 52%**

This created a binary **Meta** variable used throughout the analysis.

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## 5. Exploratory Data Analysis (EDA)

EDA was performed to identify trends, relationships, and reliable champion-level patterns.

### 5.1 Champion Win Rate vs Games Played

A scatter plot was used to analyze:

- Number of games played per champion
- Average champion win rate

This helped evaluate statistical reliability.

Champions with very low sample sizes often showed unstable or misleading win rates.

To reduce noise, only champions with **at least 500 games played** were considered reliable.

**Key Insight:**

High win rate alone was not sufficient; sample size significantly influenced reliability.

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## 5.2 Top and Bottom Champions by Win Rate

A filtered comparison of the strongest and weakest reliable champions was performed.

Only champions with at least 500 games were included.

This removed outliers caused by low-frequency champions.

**Key Insight:**

Some champions consistently outperformed others even under stricter reliability thresholds.

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## 5.3 Meta vs Non-Meta Champion Performance Across Ranks

Win rates of meta and non-meta champions were compared across all ranks.

**Selected Results:**

| Rank        | Non-Meta Win Rate | Meta Win Rate |
|-------------|-------------------|---------------|
| Bronze      | 48.9%             | 54.2%         |
| Silver      | 48.5%             | 54.2%         |
| Gold        | 49.0%             | 53.0%         |
| Emerald     | 50.1%             | 55.5%         |
| Grandmaster | 50.9%             | 58.3%         |

**Interpretation:**

Meta champions generally showed higher descriptive win rates across nearly all ranks.

This suggests a consistent performance advantage.

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## 6. Statistical Testing

To test whether meta champion advantage significantly differed between high-elo and low-elo players, a **Welch's two-sample t-test** was applied.

### **Groups:**

#### **High Elo**

- Diamond
- Master
- Grandmaster
- Challenger

#### **Low Elo**

- Bronze
- Silver
- Gold

### **Hypotheses**

**H0:** There is no significant difference in meta champion win rates between high-elo and low-elo players.

**H1:** There is a significant difference.

### **Results**

- **T-statistic = -0.221**
- **P-value = 0.825**

### **Interpretation**

Since:

**$p > 0.05$**

the null hypothesis could not be rejected.

This indicates that while meta champions generally showed higher descriptive win rates, the magnitude of this advantage did not significantly differ between high-elo and low-elo groups.

## **7. Machine Learning Analysis**

To further evaluate whether match outcomes could be predicted using available gameplay-related features, a machine learning approach was implemented.

The goal of this stage was not only to assess predictive performance, but also to determine whether meta-related variables were strong enough to explain win outcomes.

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## 7.1 Models Used

Two supervised learning models were applied:

- **Logistic Regression**
- **Random Forest Classifier**

These were selected because they represent both linear and non-linear modeling approaches.

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## 7.2 Features Used

The following features were included in prediction:

- **Meta status** (Meta / Non-meta)
- **Team side** (Blue / Red)
- **Rank**
- **Champion**
- **Game duration**

Categorical features such as rank and champion name were encoded using **One-Hot Encoding**.

A train-test split was applied:

- **80% Training Data**
  - **20% Test Data**
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## 7.3 Model Performance

### Logistic Regression

Results:

- Accuracy = **51.4%**
- Precision  $\approx$  **0.51–0.52**
- Recall  $\approx$  **0.46–0.57**

- F1-score  $\approx$  **0.48–0.54**

Interpretation:

The model performed only slightly above random guessing.

This suggests that match outcomes cannot be strongly predicted using only limited rank-based and champion-based features.

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## Random Forest

Results:

- Accuracy = **50.6%**
- Precision  $\approx$  **0.50–0.51**
- Recall  $\approx$  **0.50–0.51**
- F1-score  $\approx$  **0.50–0.51**

Interpretation:

Although Random Forest can capture non-linear relationships, performance remained close to chance level.

This suggests that additional hidden factors likely influence match outcomes.

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## 7.4 Machine Learning Interpretation

Both machine learning models showed weak predictive performance.

This indicates that while meta champions may show descriptive advantages in win rate analysis, **meta alone is not a strong standalone predictor of victory**.

League of Legends outcomes appear to depend on a broader set of interacting variables, such as:

- Team composition
  - Individual player skill
  - Communication / coordination
  - Match-specific decisions
  - In-game adaptation
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## 8. Key Findings

The main findings of this project can be summarized as follows:

- Meta champions generally showed higher descriptive win rates.
- This advantage appeared consistent across nearly all ranks.
- Statistical testing showed no significant high-elo vs low-elo difference in meta advantage.
- Small sample sizes can create misleading champion-level interpretations.
- Match outcomes are difficult to predict using only champion-based and rank-based variables.
- Machine learning models showed limited predictive power.
- Champion selection matters, but it is not the dominant determinant of victory.

Overall, the analysis suggests that meta champions may provide a practical descriptive advantage, but match outcomes are still driven by multiple interacting gameplay factors.

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## 9. Limitations

This project includes several limitations that should be considered when interpreting results.

### 9.1 Simplified Meta Definition

Meta champions were defined only using win rate threshold:

**Champion Win Rate > 52%**

This is a simplified approach.

In real competitive environments, meta can also depend on:

- Pick rate
- Ban rate
- Patch updates
- Professional play trends

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### 9.2 Missing Gameplay Variables

The dataset did not include several important variables such as:



- Team composition synergy
- Individual player mastery
- Role-specific performance
- Communication
- Tactical adaptation

These factors may significantly affect outcomes.

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### **9.3 Sampling Bias**

The dataset may not represent all League of Legends players equally.

Some ranks and champions contained substantially larger sample sizes.

This may influence descriptive findings.

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### **9.4 Correlation Does Not Imply Causation**

Higher win rates for meta champions do not necessarily mean that champion strength directly causes victory.

Some performance differences may instead reflect:

- Better player mastery
  - Selection bias
  - Strategic usage patterns
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## **10. Conclusion**

This project investigated whether meta champions provide measurable advantages in League of Legends ranked gameplay.

The analysis showed that meta champions generally demonstrated higher descriptive win rates across multiple ranks.

However, statistical testing indicated that this advantage did not significantly differ between high-elo and low-elo groups.

Additionally, machine learning models achieved only limited predictive accuracy, suggesting that champion meta alone is insufficient to explain overall match outcomes.

These findings indicate that League of Legends performance is shaped by multiple interacting factors, including:

- Champion choice
- Player skill
- Team coordination
- Match strategy
- Dynamic in-game decisions

Therefore, while meta selection may provide useful strategic advantages, it should be viewed as only one component of competitive success rather than a dominant predictor of victory.

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## **11. Future Work**

This project can be extended in several ways.

Potential improvements include:

- Patch-based meta analysis
- Role-specific champion comparisons
- Time-series analysis of meta evolution
- Inclusion of pick-rate and ban-rate metrics
- More advanced machine learning models
- Team-composition-based prediction systems
- Deep learning approaches for outcome prediction

Such improvements could provide a more realistic understanding of how competitive meta influences gameplay.

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## **Final Summary**

This project successfully combined:

- Data preparation
- Relational merging
- Feature engineering

- Exploratory data analysis
- Statistical testing
- Machine learning
- Critical interpretation

to evaluate whether champion meta influences competitive performance in League of Legends.

The findings suggest that meta contributes to performance, but victory remains a multi-factor problem.



